

Part A - Warm-up: Python basics

1. Create a new file called `lesson1_exercises.py`. Print your name, the course name and today's study goal on separate lines.
2. Create variables for a person's name, age, height in meters and whether they are currently a student. Print both the values and their types.
3. Change the value stored in one variable to a different data type. Print its type before and after the change. Explain in a comment what this demonstrates about Python.
4. Create two numeric variables and calculate addition, subtraction, multiplication, normal division, floor division, remainder and exponentiation. Print a readable label before each result.
5. Write three examples where explicit type conversion is necessary: string to int, int to float and number to string.

Part B - User input and calculations

1. Build a small profile program that asks for name and year of birth, then prints an approximate age using the current year supplied in a variable.
2. Ask the user for the price of an item and a discount percentage. Calculate and print the final price. Round the displayed result to two decimals.
3. Ask for a temperature in Celsius and convert it to Fahrenheit using $F = C * 9 / 5 + 32$.
4. Ask for the length and width of a room and calculate area and perimeter. Use descriptive variable names. Python Foundation - Lesson 1 Exercises Page 2
5. Extend one of the programs so invalid numeric input is discussed in comments: what would happen today if the user entered 'hello'? Do not solve it with exceptions yet.

Part C - Strings

1. Store a full sentence in a variable. Print its length, uppercase version, lowercase version and a version with leading/trailing whitespace removed.
2. Ask for first name and last name. Create a formatted full name using an f-string.

3. Given the string 'python programming', print the first character, last character, first six characters, last eleven characters and the entire string reversed.
4. Create a username generator: ask for first and last name, remove surrounding spaces, convert to lowercase and create a username using the first three letters of the first name plus the first five letters of the last name.
5. Given an email address, use string operations to extract the part before @ and the domain after @. Assume exactly one @ for this exercise.
6. Create a sentence containing the word 'Java'. Replace it with 'Python' and print both the original and changed sentence.

Part D - String investigation

1. Predict the output of at least eight expressions using indexing and slicing before running them. Include positive indexes, negative indexes, omitted start/end values and a step.
2. Create a variable containing 'Artificial Intelligence'. Produce at least six different slices from it and comment what each slice means.
3. Investigate the difference between .split(), .strip(), .replace() and the in operator. Write one useful example of each.
4. Demonstrate string immutability: attempt conceptually to change one character, explain why direct character assignment fails, then create a new modified string instead

Part E - Applied challenge: Registration summary

1. Build a console program that collects: first name, last name, city, year of birth and favourite programming language.
2. Normalize text input so accidental surrounding spaces do not affect the result.
3. Create a generated user ID from parts of the person's name and year of birth.
4. Print a clean multi-line summary using f-strings.
5. Print the initials, full name length excluding the space, and the favourite language reversed.

6. Add at least three extra pieces of derived information of your own choice using only concepts from Lesson 1.

Part F - Stretch challenges Python Foundation

1. Create a simple seconds converter: input total seconds and calculate whole hours, remaining minutes and remaining seconds using `//` and `%`.
2. Given a four-digit integer, extract and print each digit without converting the number to a string.
3. Create a text masking program that displays only the first two and last two characters of a supplied word, replacing the middle with `*` characters.
4. Write five short 'predict before running' examples that you could give to another student. Include at least one type conversion and two string slices.